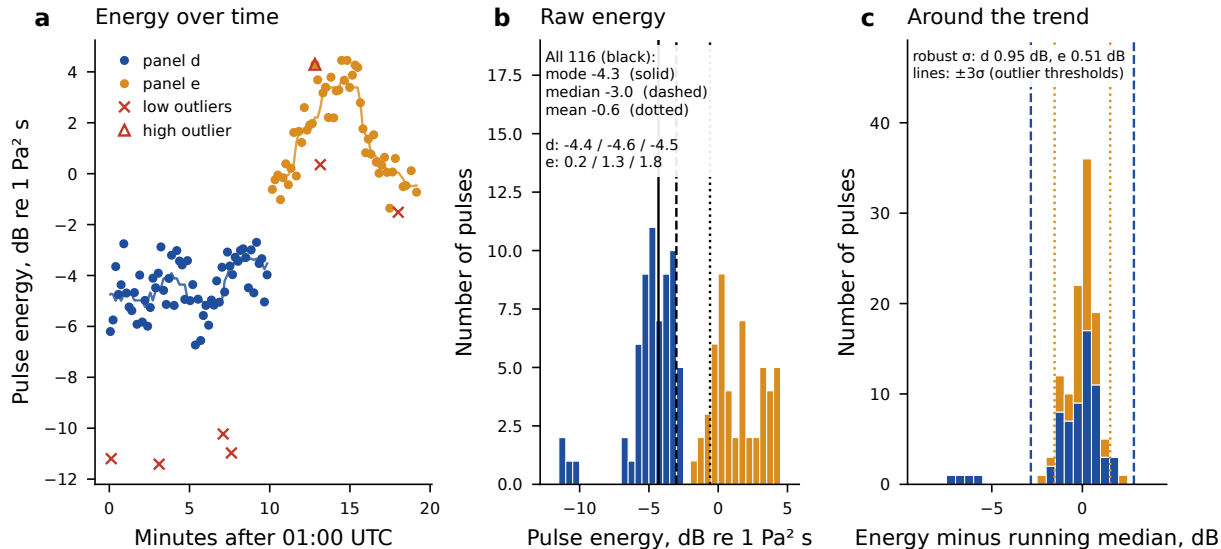




Pulse energy =  $\int p^2$  over  $\pm 1$  s about each envelope peak (prominence  $\geq 2 \times$  panel median,  $\geq 3$  s apart; no cadence rule) from Kadri (2024) Fig. 9 H08S vectors (01:00–01:20 UTC 8 Mar 2014; filtered 2–40 Hz, single channel, vertex-simplified; absolute levels approximate). (a) lines = 13-pulse running median, the trend the outlier test removes. (b) mode = KDE peak; mean = mean of linear energy in dB; values in dB re 1 Pa<sup>2</sup> s, shown as mode / median / mean for each panel. Panels d and e differ by about 6 dB (Kadri's plot scaling or survey range; not resolvable). (c) residual about the running median; outliers lie beyond  $\pm 3$  robust  $\sigma$ . Pre-registered test `h08s_pulse_energy.py` (8ff37fb); no impact-PDF input.